

ARUN SHARMA

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SUMMARY

I build physics-informed AI systems that make spatial reasoning reliable for real-world deployment, from detecting GPS spoofing to enable safer autonomous navigation, to modeling energy-efficient vehicle routing, to downscaling climate data for coastal flood risk assessment. My research fuses deep generative models (diffusion, transformers), physics-informed learning, and geostatistical priors to address data incompleteness, distribution shift, and violation of physical laws in spatial ML. CS PhD from UMN (2025). At Esri, I shipped a maritime anomaly detection pipeline on AWS that improved classification accuracy from 55% to 73% and reduced API latency by 30%. Seeking MLE and Research Scientist roles.

EDUCATION

University of Minnesota, Twin Cities – M.S. & Ph.D., Computer Science 2018 – 2025

Advisors: Prof. Shashi Shekhar | **Committee:** Prof. Vipin Kumar, Prof. Ravi Janardan, and Prof. Ying Song

Dissertation: Distortion-Aware Spatial Data Science | **Doctoral Dissertation Fellowship (2022–23)**

State University of New York at Buffalo – M.S., Computer Science 2016 – 2018

TECHNICAL SKILLS

Languages: Python, Java, R, SQL, Scala, C/C++, Julia, Rust, Go, Shell (Bash), MATLAB, JavaScript (Node.js), TypeScript.

Machine Learning: MLflow, XGBoost, MLlib, LLMs (fine-tuning & inference), VLMs, RAG, LangChain, VectorDB (FAISS, Chroma, Pinecone), Hugging Face Transformers/TRL, LoRA, PEFT, RLHF, OpenAI API, Triton Inference Server, ONNX.

Agentic AI: Chain-of-Thought (CoT), ReAct, Tool/Function Calling, Agent Traceability, Tool-Call & Token Optimization, Multi-Agent Orchestration (LangGraph, AutoGen), MCP, Prompt Caching.

Big Data & Distributed Systems: PySpark, Flink, Presto, Airflow, Kafka, RabbitMQ, Delta Lake, Docker, Terraform, Kubernetes.

ML Frameworks & Deployments: PyTorch, TensorFlow, JAX, ONNX, SageMaker, Lambda, ECS, Bedrock, vLLM/SGLang, Temporal-Nexus.

RL & Alignment: RLHF/RLAIF/RLVR, PPO, DPO, GRPO, Constrained RL, Reward Hacking Mitigation, Agent Traceability, Tool-use Safety.

EXPERIENCE

ESRI – Environmental Systems Research Institute Research Scientist Intern, May 2023 – Dec 2023

- Improved detection of illegal vessel behavior (route deviations, dark shipping) from 55% to 73% accuracy by independently designing and deploying an end-to-end anomaly detection pipeline integrating Transformer-based models, Evidential Deep Learning (EDL), and AWS SageMaker on a large-scale maritime dataset (~500M AIS records), with automated retraining via Lambda, ECS, and Step Functions.
- Reduced maritime route query latency by 40% to support real-time vessel tracking by designing and implementing a scalable Graph-based Traffic Representation and Association (GTRA) framework leveraging PySpark and GeoAnalytics APIs over a large-scale AIS maritime dataset.
- Cut model retraining time by 35% and API latency by 30% to enable faster anomaly alerting by implementing model quantization and optimizing inference with SageMaker Multi-Model Endpoints (MME), Step Functions, SQS, CloudWatch, and CI/CD pipelines.

University of Minnesota, Twin Cities Graduate Research Assistant Aug 2018 – Aug 2025

- Enabled detection of GPS spoofing for maritime safety and autonomous vehicle security by leading development of Pi-DPM, a physics-informed diffusion probabilistic model pairing a transformer Context-Informed Encoder (spatial+temporal attention over neighbor trajectories) with a physics-informed decoder embedding a single-axle Kinematic Bicycle Model (S-KBM) prior and physics regularizers on velocity, acceleration, heading, curvature, and turning rate; detected GPS-spoofed and AI deep-fake trajectories; outperforming 6 SOTA baselines (DiffWave, DiffTraj, ControlTraj) at 5/10/20% anomaly rates with strong within- and cross-domain transfer across maritime and urban domains.
- Improved coastal flood risk assessment by downscaling coarse climate projections (25km → ~1km) to fine-grained sea-level maps by co-leading a cross-disciplinary collaboration to design a Kriging-informed Conditional Diffusion Probabilistic Model, a U-Net conditional diffusion conditioned on Universal Kriging interpolations under a Matérn variogram with a variogram-based regularizer penalizing empirical-vs-observed variogram discrepancy in reverse diffusion; downscaled Copernicus CDS and CMIP6 HighResMIP (EC-Earth3P) coarse projections to fine-resolution sea-level anomaly and eddy kinetic energy maps, outperforming bicubic, CNN, GAN, and baseline diffusion on RMSE/MAE/PCC/CRPS across Eastern/Western North America.
- Built GeoTrace-Agent, a production-grade multi-agent framework for auditable spatiotemporal reasoning over heterogeneous trajectory and Earth-observation sources, by designing a typed PlanGraph chain-of-thought planner that decomposes natural-language spatial queries into deterministic sub-tasks over AIS feeds, OSM road networks, Copernicus weather, Sentinel imagery, and space-time-prism tools; integrated specialized agents for Hägerstrand prism reasoning, STAGD-DRM abnormal trajectory-gap detection, TGARD/DC-TGARD rendezvous discovery, and S-KBM kinematic validation, exposed through MCP and JSON-RPC A2A protocols with OpenTelemetry tracing, Postgres HITL review queues, semantic caching, tool-call deduplication, and adaptive prompt compression; reduced per-query token spend by ~40%, lowered cost from ~\$0.054 to ~\$0.034/query versus no-cache ablations, and enforced physical feasibility by rejecting infeasible regions before user-facing output.
- Designed Pi-GRPO, a physics-informed reinforcement-learning stack for trajectory generation and trajectory-reasoning policies, by unifying PPO, DPO, and GRPO under a shared hybrid reward composed of an unbounded hard S-KBM violation penalty, a 95th-percentile jerk/curvature soft envelope, a Pi-DPM reconstruction-likelihood term, and an optional cross-encoder preference model; implemented GRPO group-baseline advantages without a value head, DPO with a physics-aware γ_{phys} augmentation, PPO with bounded adaptive KL control, vLLM-backed online rollouts with prefix caching, content-addressed checkpoints, W&B/OpenTelemetry reward decomposition, and HITL-to-DPO preference-data curation from GeoTrace-Agent; trained on ~11K preference triples and demonstrated reward-hacking resistance by reducing hard-violation rate from 18% under vanilla DPO to 0% with physics-augmented DPO, while maintaining bounded KL behavior across 3,000 PPO steps.

SELECTED PROJECTS

- Generated realistic synthetic vehicle trajectories for autonomous driving simulation and traffic prediction by co-designing GCDM (Geo-lucid Conditional Diffusion Model), a two-stage cascaded generative diffusion injecting attributed road-network priors via map-informed latent variables, spatial cross-attention, and a physics-based trajectory decomposition (map-derived base + learned residual); produced trajectories with stronger geo-distribution similarity (density, trip error) and higher velocity/acceleration/jerk fidelity than 4 baselines (DiffTraj, Sashimi, DiffWave, EETG) on Porto (~1.7M trips) and Harbin taxi data, while reducing downstream traffic in/out-flow prediction error.
- Enabled real-time energy-vs-travel-time route planning for electric vehicles by co-designing MBOR (Multi-Level Bi-Objective Routing), introducing three novel data structures, a boundary multigraph, Multi-level Encoded Pareto Frontier View (MEPFV), and 2D cost-interval pruning with multi-edge pruning, to precompute and retrieve complete Pareto-optimal path sets over fragmented road networks; achieved >10× online-runtime improvement on real-world road network data for real-time bi-objective EV route planning.
- Reduced manual investigation area for illegal maritime activity (fishing, oil transfers, trans-shipments) by up to 80% by leading a multi-year, 4-publication research program developing STAGD (Space-Time Aware Gap Detection) with a Dynamic Region Merge (DRM) criterion combining comparison-less temporal indexing, R*-tree hierarchical spatial indexing, and maximal-union merge over space-time prism geo-ellipses; proposed TGARD and Dual Convergence DC-TGARD exploiting ellipse symmetry with bi-directional pruning and early-stopping; experimentally faster and more accurate than baselines AIS data (500K ships) for homeland-security and public-safety applications.
- Accelerating computationally expensive ecosystem models for sustainable agriculture: developed SM-Hybrid, a novel surrogate model for the Daycent agroecosystem model capable of explicitly modeling spatial autocorrelation (within-site dependencies) and tele-connections (cross-site long-range dependencies) using hybrid spatial neural network architectures. Predicted land emissions (GPP, Ra, Rh, NEE, crop yield) across Iowa/Illinois/Indiana 2× faster than Daycent while achieving higher accuracy than SM-ANN baselines, enabling rapid scenario evaluation for climate-smart agricultural policy.
- Enabling precision immunotherapy through spatial analysis of tumor microenvironment: developed domain-adapted spatially-lucid neural networks for multi-category point set classification in non-Euclidean space, using a spatial ensemble framework with place-calibration parameters, weighted-distance learning rate, and spatial domain adaptation across heterogeneous tissue place-types. Leveraged a multi-task architecture with spatial mix-up masking and spatial contrastive predictive coding for self-supervised learning on MxIF oncology data to classify immune-tumor relationships for immunotherapy.
- Community-engaged spatial epidemiology for pandemic response: built an entity-relationship model, system architecture, and implementation to analyze aggregated mobile device data for understanding COVID-19 policy interventions on mobility. Delivered county-level analytics on long-duration visits to high-risk business categories and fine-resolution device count maps to epidemiologists, analysts, and policymakers. Collaborated directly with the Minnesota Department of Management and Budget; work cited in testimony to the MN House Transportation Finance Committee.

HONORS AND ACHIEVEMENTS

DOCTORAL DISSERTATION FELLOWSHIP

2022 - 2023

University of Minnesota, Twin Cities

TEACHING EXPERIENCE

SPATIAL DATA SCIENCE RESEARCH

Spring 2024

- Guest lectures on scientific methodology, fostering research skills, and effectively communicating research ideas.
- Supervised diverse projects and mentored multiple graduate and undergraduate students from interdisciplinary fields.

SPATIAL DATA SCIENCE

Fall 2019

- Responsible for handling class lectures, queries, homework assignments, labs, exams, lecture slides, etc.
- Guest lecture topics: Spatial Indexing, Networks, and Data Mining.

ADVANCED DATABASE SYSTEMS

Spring 2019

- Responsible for handling class lectures, queries, homework assignments, labs, exams, lecture slides, etc.
- Guest lecture topics: Concurrency Control, Database Security, and Data Mining.

DATA STRUCTURES AND ALGORITHMS

Fall 2018

- Instructed weekly recitation sessions with over 40+ students and graded 400+ students.

SERVICES AND LEADERSHIP

SUST 4096 SUSTAINABILITY INTERNSHIP, UNIVERSITY OF MINNESOTA, MN (Mentor)

Spring 2024

- Supervised an undergraduate student on a course project based on KGML for Sustainable and Precision Agriculture.
- Presented a lightning talk and a poster at the Annual AI-LEAF PI Meeting in UMN Saint Paul Campus, MN¹.

HONORS MENTORS CONNECTION, WAYZATA HIGH SCHOOL, MN (Mentor)

Fall 2023 - Spring 2024

- Advised a high school student to analyze a real-world case study of signal spoofing behavior in the maritime domain.
- The student helped analyze circular patterns, enabling us to create a taxonomy of spoofing behavior in open waters.

MINNESOTA DEPARTMENT OF MANAGEMENT AND BUDGET, MN (Service)

May 2020 - June 2021

- Reported county-level mobility traffic to epidemiologists, analysts, and policymakers for informed decision-making.²
- Advised multiple high school and undergraduate students who were considering a research career.

¹ <https://cse.umn.edu/aiclimat/news/2024-ai-climate-annual-review-meeting>

² Future of transportation in a post-pandemic world, Prof. Shashi Shekhar, Transportation Finance and Policy Committee, Minnesota House of Representatives, Jan. 14th, 2021.

INVITED PRESENTATIONS AND POSTERS

Poster: Towards Physics-guided Generative Foundation Models The 1st ACM SIGSPATIAL International Workshop on Generative and Agentic AI for Multi-Modality Space-Time. Authors: Arun Sharma, Majid Farhadloo, Mingzhou Yang, Bharat Jayaprakash, William Northrop, and Shashi Shekhar	2025
Poster: Spatial Distribution-Shift Aware Knowledge-Guided Machine Learning AAAI Bridge on Knowledge-Guided ML: Bridging Scientific Knowledge and AI, 2024. Authors: Arun Sharma, Majid Farhadloo, Mingzhou Yang, Ruolei Zeng, Subhankar Ghosh, and Shashi Shekhar	2024
Presentation: Towards Distortion Aware Spatial Data Science Michigan Institute of Data Science and Society (MIDAS), University of Michigan, Ann Arbor Authors: Arun Sharma and Shashi Shekhar	2023

SELECTED PUBLICATIONS

[1] Towards Physics-informed Diffusion for Anomaly Detection in Trajectories: A Summary of Results
Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection (GeoAnomalies '25)
Arun Sharma, Mingzhou Yang, Majid Farhadloo, Subhankar Ghosh, Bharat Jayaprakash, and Shashi Shekhar

[2] Towards Kriging-informed Conditional Diffusion for Regional Sea-Level Data Downscaling: A Summary of Results
32nd International Conference on Advances in Geographic Information Systems (SIGSPATIAL '24)
Subhankar Ghosh*, Arun Sharma*, Jayant Gupta, Aneesh Subramanian and Shashi Shekhar (*Both authors contributed equally to this paper)

[3] Physics-based Abnormal Trajectory-Gap Detection
ACM Transactions in Intelligent Systems and Technology, 2024.
Arun Sharma, Subhankar Ghosh, and Shashi Shekhar

[4] Analyzing Trajectory Gaps for Possible Rendezvous Regions
ACM Transactions in Intelligent Systems and Technology, 2022
Arun Sharma and Shashi Shekhar

[5] Towards a Tighter Bound on Possible-Rendezvous Areas: Preliminary Results
30th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '22)
Arun Sharma, Jayant Gupta, Subhankar Ghosh, and Shashi Shekhar

[6] Geo-lucid Conditional Diffusion Models for High Physical Fidelity Trajectory Generation
Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25)
Mingzhou Yang, Arun Sharma, Majid Farhadloo, Bharat Jayaprakash, and Shashi Shekhar

[7] Towards Surrogate Models with Hybrid Spatial Neural Networks: A Summary of Results
Proceedings of the 8th ACM SIGSPATIAL International Workshop on Geospatial Simulation (GeoSIM '25), pp. 57–69, 2025
Shengya Zhang, Arun Sharma, Majid Farhadloo, Mingzhou Yang, Yao Zhang, Mu Hong, Licheng Liu, David Mulla, and Shashi Shekhar

[8] Towards Fine-Tuning-Based Site Calibration for Knowledge-Guided Machine Learning: A Summary of Results
5th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)
Ruolei Zeng, Arun Sharma, Shuai An, Mingzhou Yang, Shengya Zhang, Licheng Liu, David Mulla, and Shashi Shekhar

[9] Spatially-Delineated Domain-Adapted AI Classification: An Application for Oncology Data
SIAM International Conference on Data Mining (SIAM SDM) 2025
Majid Farhadloo, Arun Sharma, Alexey Leontovich, Svetomir N. Markovic, and Shashi Shekhar

[10] Towards Spatially-Lucid AI Classification in Non-Euclidean Space: An Application for MxIF Oncology Data
SIAM International Conference on Data Mining (SIAM SDM) 2024
Majid Farhadloo, Arun Sharma, Jayant Gupta, Alexey Leontovich, Svetomir N. Markovic, and Shashi Shekhar

[11] Spatiotemporal Data Mining: A Survey
Handbook of Spatial Analysis for the Social Sciences, Edward Elgar, 2022
Arun Sharma, Zhe Jiang, and Shashi Shekhar

[12] Understanding COVID-19 effects on mobility: A community-engaged approach
25th AGILE Conference on Geographic Information Science, 2022
Arun Sharma, Majid Farhadloo, Yan Li, Jayant Gupta, Aditya Kulkarni, and Shashi Shekhar

[13] WebGlobe: A cloud-based framework for interacting with climate data
International Workshop on Analytics for Big Geospatial Data (SIGSPATIAL) 2018
Arun Sharma, SM Arshad Zaidi, Varun Chandola, Melissa R Dumas, and Budhendra L Bhaduri

REVIEWER: NeurIPS, ICML, ICLR, CVPR, ECCV, AAAI, IJCAI, SIGKDD, CIKM, ICDM, SDM, MLsys, SIGSPATIAL, TKDE, JMLR

REFERENCES (CONTACT INFORMATION)

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